

F866 — the map make-or-break (my lead), honest first computation: ONE solid reduction + ONE identified obstacle + NO map claim (I ran the numbers and the easy map FAILED). The credential sentence — “your fermionic projector is the Bergman kernel of this bounded domain” — is NOT earned yet, and I will not say it until the math says it. Two results from actually computing, per the phantom lesson: (1) SOLID reduction — the scalar Bergman amplitude factors OUT of the causal classification. The Bergman kernel is Hermitian-symmetric ($K(y,x) = \text{conj}(K(x,y))$), so $K(x,y)K(y,x) = |K(x,y)|^2 \geq 0$ (real, positive). For a covariant kernel $P(x,y) = K(x,y) \cdot \Sigma(x-y)$ (scalar amplitude $\times$ spinor structure), the closed chain $A_{xy} = P(x,y)P(y,x) = |K|^2 \cdot [\Sigma(x-y)\Sigma(y-x)]$, and since $|K|^2$ is real-positive, the eigenvalue REALITY-vs-COMPLEXITY (= Def 1.3’s timelike-vs-spacelike) comes ENTIRELY from the (2,2) SPINOR factor Σ , not the amplitude. This is genuine and target-innocent: causality lives in the spinor structure. (2) OBSTACLE identified by FAILURE — I ran the spinor closed chain for naive kernels and it does NOT reproduce Def 1.3: for $P = a \cdot I + b \cdot (\gamma \cdot \xi)$ with the ξ -reversed $P(y,x)$, the Clifford identity $(\gamma \cdot \xi)^2 = \xi^2$ collapses $A_{xy} = (a^2 - b^2 \xi^2) \cdot I \rightarrow \text{ALL-REAL}$ for both timelike AND spacelike (wrong); forcing the $i\epsilon$ /complex coefficients flips the classification BACKWARDS (timelike-complex, spacelike-real). The reason: in CFS the closed chain uses $P(y,x) =$ the (2,2)-SPIN ADJOINT of $P(x,y)$ ($P^\dagger = S^{-1}P^\dagger S$ w.r.t. the indefinite (2,2) spin form S), NOT the ξ -reversed kernel I was substituting, and NOT the standard Dirac (1,3) gammas — the correct structure is the conformal (2,2) spin ($SU(2,2)$) with its indefinite adjoint. So the make-or-break hinges on the EXACT (2,2)-adjoint fermionic-projector structure, which my naive constructions get wrong. HONEST STATE: the map is NOT shown — and I demonstrated (by computing) that it is not a freebie; the classification depends on the exact (2,2)-spin adjoint structure of the kernel, which needs Finster’s exact P construction (from the source) or our exact (2,2)-spinor Bergman-kernel boundary limit, computed carefully. NO map claim; “IF the map holds, then...”; Cal’s ceiling (continuum-limit leading-order, never forces Einstein) intact. This is the phantom lesson working a second time: I ran the number, the easy map failed, and I report the obstacle rather than the beautiful sentence.

Lyra, Saturday 2026-08-08 (post-break). The make-or-break map, computed honestly. Delivered the scalar reduction (solid) + the identified obstacle (the (2,2)-adjoint structure) + NO claim, because the numbers refused the easy map twice. Decided on the math, blind. RUBRICS done-bar below.

What I computed (run the number)

construction	closed chain A_{xy}	vs Def 1.3
$P = a \cdot I + b \cdot (\gamma \cdot \xi)$, $P(y,x) = \xi$ -reversed	$(a^2 - b^2 \xi^2) \cdot I \rightarrow$ all real (both)	WRONG (real for spacelike too)
same, complex ($i\varepsilon$) coefficients	timelike $\rightarrow$ complex, spacelike $\rightarrow$ real	BACKWARDS
correct object	$P(y,x) = (2,2)$ -spin adjoint $P^\dagger = S^{-1} P^\dagger S$ (conformal $SU(2,2)$, indefinite S)	the real test — NOT yet computed

The two honest deliverables

1. SOLID (scalar reduction): causality is in the (2,2) spinor factor; the scalar Bergman amplitude $|K|^2$ factors out (real-positive). Target-innocent. This simplifies the map to a spinor question.
2. OBSTACLE (from failure): the classification hinges on the (2,2)-spin ADJOINT structure ($P(y,x) = P^\dagger$, indefinite S), NOT ξ -reversal or standard Dirac gammas. My naive kernels fail; the exact conformal-(2,2)-spin fermionic projector is required.

RUBRICS Layer-2 done-bar (make-or-break, honest)

- ☒ Mechanism concrete: $A_{xy} = |K|^2 \cdot \Sigma(x) \Sigma(y)$; causality in Σ (the (2,2) spinor); test = Def 1.3 on Σ 's closed chain.
- ☒ Open edge named precisely: the (2,2)-spin adjoint structure $P(y,x) = S^{-1} P^\dagger S$ — get from source or the (2,2)-spinor Bergman boundary limit.
- ☒ Tier = open/Framework; map NOT shown; “if the map holds”; Einstein NOT promoted (Cal ceiling).
- ☒ Ran the number before claiming; the easy map FAILED; reported the obstacle, not a headline.

Handoffs

- @Keeper — the map's make-or-break, honest: the scalar Bergman amplitude factors OUT (causality is purely in the (2,2) spinor structure — a real reduction), but my computed tests of naive kernels FAIL Def 1.3 (real-for-both, or backwards). The obstacle is precise: the closed chain uses the (2,2)-spin ADJOINT $P(y,x) = S^{-1} P^\dagger S$ (conformal $SU(2,2)$), not the ξ -reversed kernel. So the credential sentence is NOT earned; the next step is the exact (2,2)-spin fermionic projector (source or the Bergman boundary limit). Hold the Finster note — the map has not landed.
- @Elie — if your CFS calculator constructs $P(y,x)$, check whether it uses the (2,2)-spin adjoint ($S^{-1} P^\dagger S$) or ξ -reversal — my failures came from the latter. The correct object is the $SU(2,2)$ conformal-spin adjoint.
- @Cal — guard held: NO map claim; I ran the numbers, the easy map failed twice, and I reported the obstacle. The ceiling (continuum-limit leading-order, never Einstein) is untouched because there is no match to build on yet.
- @Casey — the beautiful sentence isn't earned yet, and I want to be straight that I tried the honest way and the math said “not so fast.” One real thing did fall out: the *size* part of the Bergman kernel — the raw amplitude — drops out of the question entirely; whether two points are space- or time-separated is decided purely by the *spinor* part, the little internal arrow each point carries. That's a genuine simplification, and it's the part a geometer would nod at. But when I actually computed the causal test with the obvious form of that spinor part, it gave the wrong answer — the same for space and time — and a quick fix flipped it backwards. The reason is subtle and I pinned it: the “reverse” point isn't just the coordinate reversed, it's an *adjoint* taken with respect to the funny (2,2)-signature inner product our spin space carries — and I was using the ordinary one. So the map isn't wrong, but it isn't a freebie either; it needs the exact form of that adjoint, which I'll get from Finster's construction rather than guess. No claim, no Finster note yet — decided on the math, blind, exactly as you said. Nothing pushed.

Notes only; no toy/theorem claimed (honest make-or-break state). F866: the Bergman-kernel= $P(x,y)$ map, first computation. SOLID: scalar amplitude factors out ($K(x,y)K(y,x)=|K|^2$ real-positive) $\rightarrow$ causality is entirely in the (2,2) spinor factor. FAILED (ran the number): naive $P=a\cdot I+b\cdot \gamma\xi$ with ξ -reversed $P(y,x) \rightarrow A_{xy}=(a^2-b^2\xi^2)\cdot I$ all-real for both (wrong); complex coeffs $\rightarrow$ backwards. OBSTACLE: the closed chain needs $P(y,x) =$ the (2,2)-SPIN ADJOINT $S^{-1}P^\dagger S$ (conformal $SU(2,2)$, indefinite S), NOT ξ -reversal or standard Dirac gammas. Map NOT shown; needs the exact (2,2)-spin fermionic projector (source / Bergman boundary limit). NO claim, if-then, Cal ceiling intact. Phantom lesson worked again: ran it, easy map failed, reported the obstacle. @Elie check your $P(y,x) =$ adjoint vs ξ -reversal; @Keeper hold the Finster note. — Lyra